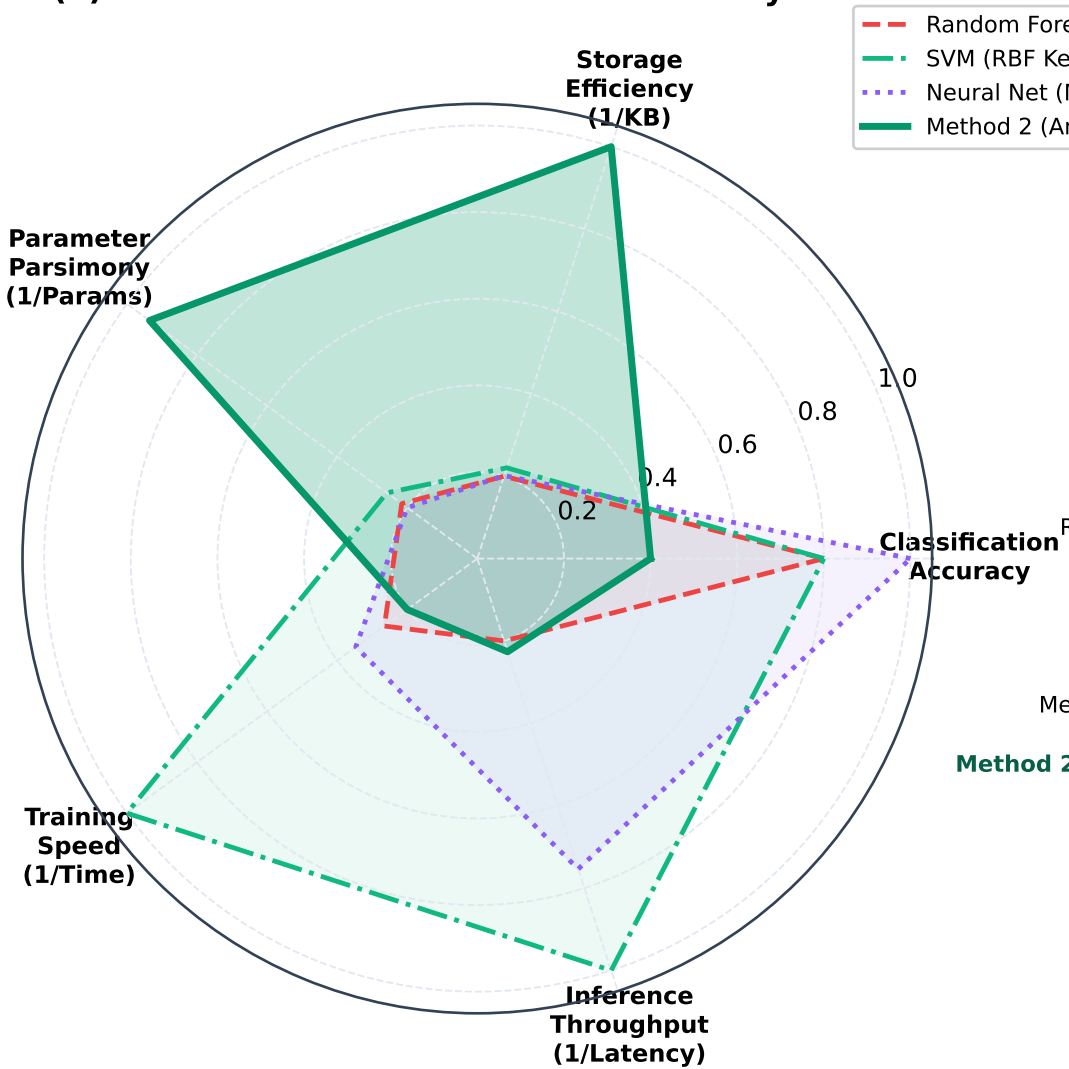
Comparative Computational Complexity: Space, Time, and Memory Scaling Across Models

(a) Multi-Criteria Architectural Efficiency Radar



(b) Empirical Time, Space, and Asymptotic Complexity Benchmark

Model Architecture	Params	Disk Size	Train Time	Latency	Test Acc	Space Complexit
Logistic Regression (SGD)	32	1.12 KB	1 ms	0.06 ms	93.94%	O(D)
Support Vector Machine (RBF)	1,921	16.87 KB	2 ms	0.05 ms	90.91%	O(N_sv · D)
Random Forest Ensemble (150 Tre	3,328	302.24 KB	142 ms	12.86 ms	90.91%	O(T · Nodes)
XGBoost Gradient Boosting	682	71.94 KB	69 ms	0.15 ms	93.94%	O(T · Depth)
Multi-Layer Perceptron (64-32)	4,161	135.46 KB	19 ms	0.09 ms	93.94%	O(H_1 · H_2)
Method 1: Angle-Encoded VQC (PCA	38	0.45 KB	48.20 s	6.62 ms	81.82%	O(n) Qubits
Method 2: Amplitude-Encoded VQC (H	**38**	**0.45 KB**	**52.40 s**	**6.29 ms**	**84.85%**	**O(log_2 D) Qubits**